

1. Mostrar la estructura del sistema

ls /

```
jcsantos@JCSantos-Ubunut:~  
└─ jcsantos at JCSantos-Ubunut in ~  
    └─ ls /  
        bin  cdrom  etc  lib  lost+found  mnt  proc  run  snap  sys  usr  
        boot dev  home lib64 media  opt  root  sbin  srv  tmp  var  
└─ jcsantos at JCSantos-Ubunut in ~  
    └─
```

2. Ver contenido de /etc

ls -l /etc

```
jcsantos@JCSantos-Ubunut:~  
└─ jcsantos at JCSantos-Ubunut in ~  
    └─ ls -l /etc  
        total 1180  
        drwxr-xr-x  2 root          root          4096 Jan  8 09:37 ImageMagick-7  
        drwxr-xr-x  5 root          root          4096 Oct  7 04:35 ModemManager  
        drwxr-xr-x  8 root          root          4096 Jan 16 14:14 NetworkManager  
        drwxr-xr-x  2 root          root          4096 Jan  8 10:03 PackageKit  
        drwxr-xr-x  2 root          root          4096 Oct  7 04:36 UPower  
        drwxr-xr-x 11 root          root          4096 Oct  7 04:38 X11
```

3. Listar todos los dispositivos de bloque

lsblk

```
jcsantos@JCSantos-Ubunut:~  
└─ jcsantos at JCSantos-Ubunut in ~  
    └─ lsblk  
NAME    MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS  
fd0      2:0    1    4K  0 disk  
loop0    7:0    0    4K  1 loop /snap/bare/5  
loop1    7:1    0  73.9M  1 loop /snap/core22/2216  
loop2    7:2    0  73.9M  1 loop /snap/core22/2133  
loop3    7:3    0  11.8M  1 loop /snap/desktop-security-center/59  
loop4    7:4    0  22.6M  1 loop /snap/desktop-security-center/99  
loop5    7:5    0 247.6M  1 loop /snap/firefox/6966  
loop6    7:6    0  18.5M  1 loop /snap/firmware-updater/210  
loop7    7:7    0  11.1M  1 loop /snap/firmware-updater/167  
loop8    7:8    0 516.2M  1 loop /snap/gnome-42-2204/226  
loop9    7:9    0 531.4M  1 loop /snap/gnome-42-2204/247  
loop10   7:10   0  91.7M  1 loop /snap/gtk-common-themes/1535
```

4. Mostrar los inodos de varios archivos

ls -li /etc/passwd

```
jcsantos@JCSantos-Ubunut:~  
jcsantos at JCSantos-Ubunut in ~  
ls -li /etc/passwd  
1056548 /etc/passwd  
jcsantos at JCSantos-Ubunut in ~
```

ls -li /var/log/syslog

```
jcsantos@JCSantos-Ubunut:~  
jcsantos at JCSantos-Ubunut in ~  
ls -li /var/log/syslog  
1055101 /var/log/syslog  
jcsantos at JCSantos-Ubunut in ~
```

5. Ver información completa de un archivo

stat /etc/hosts

```
jcsantos@JCSantos-Ubunut:~  
jcsantos at JCSantos-Ubunut in ~  
stat /etc/passwd  
File: /etc/passwd  
size: 2911      Blocks: 8      IO Block: 4096  regular file  
Device: 802h/2050d    Inode: 1056548  Links: 1  
Access: (0644/-rw-r--r--)  Uid: (  0/   root)   Gid: (  0/   root)  
Access: 2026-01-19 11:29:01.149652213 +0100  
Modify: 2026-01-19 11:29:00.552489080 +0100  
Change: 2026-01-19 11:29:00.553315384 +0100  
Birth: 2026-01-19 11:29:00.551350518 +0100  
jcsantos at JCSantos-Ubunut in ~
```

6. Localizar dónde está montado un USB

df -h

```
jcsantos@JCSantos-Ubunut:~  
jcsantos at JCSantos-Ubunut in ~  
df -h  
Filesystem      Size  Used Avail Use% Mounted on  
tmpfs           1.5G  2.1M  1.5G   1% /run  
/dev/sda2       20G   7.1G  12G  39% /  
tmpfs           3.7G   0  3.7G   0% /dev/shm  
tmpfs           5.0M  8.0K  5.0M   1% /run/lock  
tmpfs           3.7G  8.0K  3.7G   1% /tmp  
tmpfs           1.0M   0  1.0M   0% /run/credentials/systemd-journald.service  
tmpfs           1.0M   0  1.0M   0% /run/credentials/systemd-resolved.service  
tmpfs          739M   88K  739M   1% /run/user/1000  
/dev/sdb1       4.4G  3.5M  4.3G   1% /media/jcsantos/JCSantos  
jcsantos at JCSantos-Ubunut in ~
```

7. Crear un directorio para montajes manuales

`sudo mkdir /mnt/usbtest`

```
jcsantos@JCSantos-Ubunut:~  
jcsantos at JCSantos-Ubunut in ~  
└─○ sudo mkdir /mnt/usbmount  
[sudo: authenticate] Password:  
jcsantos at JCSantos-Ubunut in ~  
└─○
```

8. Cambiar permisos a un directorio

`sudo chmod 755 /mnt/usbtest`

```
jcsantos@JCSantos-Ubunut:~  
jcsantos at JCSantos-Ubunut in ~  
└─○ sudo mount /dev/sdb1 /mnt/usbmount  
mount: /mnt/usbmount: /dev/sdb1 already mounted on /mnt/usbmount.  
dmesg(1) may have more information after failed mount system call.  
jcsantos at JCSantos-Ubunut in ~  
└─○ df -h  
Filesystem      Size  Used Avail Use% Mounted on  
tmpfs           1.5G  2.1M  1.5G   1% /run  
/dev/sda2       20G   7.1G  12G   39% /  
tmpfs           3.7G   0  3.7G   0% /dev/shm  
tmpfs           5.0M   8.0K  5.0M   1% /run/lock  
tmpfs           3.7G   8.0K  3.7G   1% /tmp  
tmpfs           1.0M   0  1.0M   0% /run/credentials/systemd-journald.service  
tmpfs           1.0M   0  1.0M   0% /run/credentials/systemd-resolved.service  
tmpfs          739M   88K  739M   1% /run/user/1000  
/dev/sdb1       4.4G   3.5M  4.3G   1% /mnt/usbmount  
jcsantos at JCSantos-Ubunut in ~  
└─○
```

9. Identificar tu sistema de archivos principal

`df -T /`

```
jcsantos@JCSantos-Ubunut:~  
jcsantos at JCSantos-Ubunut in ~  
└─○ df -T /  
Filesystem      Type 1K-blocks    Used Available Use% Mounted on  
/dev/sda2       ext4 20463184 7418236 11980144  39% /  
jcsantos at JCSantos-Ubunut in ~  
└─○
```

10. Ver logs recientes del sistema

sudo tail /var/log/syslog

```
jcsantos@JCSantos-Ubunut:~  
jcsantos at JCSantos-Ubunut in ~  
└─○ sudo tail /var/log/syslog  
2026-01-19T13:11:16.017554+01:00 JCSantos-Ubunut ubuntu-insights[3630]: }  
2026-01-19T13:11:16.017623+01:00 JCSantos-Ubunut ubuntu-insights[3630]: 2026/01/19 13:11:16 ERROR failed to  
collect insights: insights write failed: report already exists for this period  
2026-01-19T13:11:16.018126+01:00 JCSantos-Ubunut systemd[1884]: Finished ubuntu-insights-collect.service -  
"Collect platform report using Ubuntu Insights while respecting consent".  
2026-01-19T13:13:48.755219+01:00 JCSantos-Ubunut systemd[1884]: Starting ubuntu-insights-upload.service -  
"Upload collected and matured platform reports using Ubuntu Insights while respecting consent"...  
2026-01-19T13:13:48.769640+01:00 JCSantos-Ubunut kernel: audit: type=1400 audit(1768824828.768:248): appar  
mor="AUDIT" operation="usersns_create" class="namespace" info="Usersns create - transitioning profile" profi  
le="unconfined" pid=3723 comm="(insights)" requested="usersns_create" target="unprivileged_usersns" execpath  
="/usr/lib/systemd/systemd-executor"  
2026-01-19T13:13:48.770749+01:00 JCSantos-Ubunut kernel: audit: type=1400 audit(1768824828.769:249): appar  
mor="DENIED" operation="capable" class="cap" profile="unprivileged_usersns" pid=3723 comm="(insights)" capa  
bility=21 capname="sys_admin"  
2026-01-19T13:13:48.783692+01:00 JCSantos-Ubunut ubuntu-insights[3723]: 2026/01/19 13:13:48 WARN Failed to  
upload report file=1767863030.json source=ubuntu_desktop_provision error="report send failed\nfailed to s  
end HTTP request: Post \"https://metrics.ubuntu.com/upload/ubuntu_desktop_provision\": dial tcp: lookup me  
trics.ubuntu.com on 127.0.0.53:53: dial udp 127.0.0.53:53: socket: address family not supported by protoco  
l"  
2026-01-19T13:13:48.783789+01:00 JCSantos-Ubunut ubuntu-insights[3723]: 2026/01/19 13:13:48 WARN Failed to  
send report, retrying upload after backoff period seconds=23.767133267 error="ubuntu_desktop_provision up  
load failed for report 1767863030.json: report send failed\nfailed to send HTTP request: Post \"https://me  
trics.ubuntu.com/upload/ubuntu_desktop_provision\": dial tcp: lookup metrics.ubuntu.com on 127.0.0.53:53:  
dial udp 127.0.0.53:53: socket: address family not supported by protocol"  
2026-01-19T13:13:48.783814+01:00 JCSantos-Ubunut ubuntu-insights[3723]: 2026/01/19 13:13:48 WARN Failed to  
upload report file=1767863366.json source=linux error="report send failed\nfailed to send HTTP request: P  
ost \"https://metrics.ubuntu.com/upload/linux\": dial tcp: lookup metrics.ubuntu.com on 127.0.0.53:53: dia  
l udp 127.0.0.53:53: socket: address family not supported by protocol"  
2026-01-19T13:13:48.783849+01:00 JCSantos-Ubunut ubuntu-insights[3723]: 2026/01/19 13:13:48 WARN Failed to  
send report, retrying upload after backoff period seconds=29.465651076 error="linux upload failed for rep  
ort 1767863366.json: report send failed\nfailed to send HTTP request: Post \"https://metrics.ubuntu.com/up  
load/linux\": dial tcp: lookup metrics.ubuntu.com on 127.0.0.53:53: dial udp 127.0.0.53:53: socket: addres  
s family not supported by protocol"  
jcsantos at JCSantos-Ubunut in ~  
└─○
```

✓ EJERCICIOS INTERMEDIOS (11–20)

11. Montar un dispositivo USB manualmente

```
sudo mount /dev/sdb1 /mnt/usbtest
```

12. Desmontarlo

```
sudo umount /mnt/usbtest
```

```
jcsantos@JCSantos-Ubuntu:~  
└─ jcsantos at JCSantos-Ubuntu in ~  
  └─ sudo mount /dev/sdb1 /mnt/usbmount  
  └─ jcsantos at JCSantos-Ubuntu in ~  
    └─ lsblk  
NAME      MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS  
fd0        2:0    1    4K  0 disk  
loop0      7:0     0    4K  1 loop /snap/bare/5  
loop1      7:1     0   73.9M 1 loop /snap/core22/2133  
loop2      7:2     0   73.9M 1 loop /snap/core22/2216  
loop3      7:3     0   11.8M 1 loop /snap/desktop-security-center/59  
loop4      7:4     0   22.6M 1 loop /snap/desktop-security-center/99  
loop5      7:5     0  247.6M 1 loop /snap/firefox/6966  
loop6      7:6     0  251.6M 1 loop /snap/firefox/7672  
loop7      7:7     0   11.1M 1 loop /snap/firmware-updater/167  
loop8      7:8     0   18.5M 1 loop /snap/firmware-updater/210  
loop9      7:9     0   91.7M 1 loop /snap/gtk-common-themes/1535  
loop10     7:10    0  516.2M 1 loop /snap/gnome-42-2204/226  
loop11     7:11    0  531.4M 1 loop /snap/gnome-42-2204/247  
loop12     7:12    0   14.4M 1 loop /snap/prompting-client/104  
loop13     7:13    0   10.8M 1 loop /snap/snap-store/1270  
loop14     7:14    0   17.5M 1 loop /snap/snap-store/1300  
loop15     7:15    0   48.1M 1 loop /snap/snapd/25935  
loop16     7:16    0   50.8M 1 loop /snap/snapd/25202  
loop17     7:17    0    576K 1 loop /snap/snapd-desktop-integration/315  
sda        8:0     0   20G  0 disk  
└─ sda1     8:1     0    1M  0 part  
└─ sda2     8:2     0   20G  0 part /  
sdb        8:16    1   7.4G  0 disk  
└─ sdb1     8:17    1   4.3G  0 part /mnt/usbmount  
└─ sdb2     8:18    1    3G  0 part  
└─ sdb3     8:19    1   48M  0 part  
└─ jcsantos at JCSantos-Ubuntu in ~  
  └─ sudo umount /mnt/usbmount  
  └─ jcsantos at JCSantos-Ubuntu in ~  
    └─
```

13. Ver UUID de todas las particiones

blkid

```
jcsantos@JCSantos-Ubunut:~  
jcsantos at JCSantos-Ubunut in ~  
└─○ sudo blkid  
/dev/loop1: BLOCK_SIZE="131072" TYPE="squashfs"  
/dev/loop17: BLOCK_SIZE="131072" TYPE="squashfs"  
/dev/loop8: BLOCK_SIZE="131072" TYPE="squashfs"  
/dev/sdb2: UUID="c78898c0-59a2-44b1-ad2b-af4a072f899a" BLOCK_SIZE="4096" TYPE="ext4" PARTUUID="a44a206c-02"  
/dev/sdb3: UUID="a886e47b-afe2-407f-b3a8-c697a4b76cd6" BLOCK_SIZE="4096" TYPE="ext4" PARTUUID="a44a206c-03"  
/dev/sdb1: LABEL="JCSantos" UUID="696E1D16" BLOCK_SIZE="512" TYPE="exfat" PARTUUID="a44a206c-01"  
/dev/loop15: BLOCK_SIZE="131072" TYPE="squashfs"  
/dev/loop6: BLOCK_SIZE="131072" TYPE="squashfs"  
/dev/loop13: BLOCK_SIZE="131072" TYPE="squashfs"  
/dev/loop4: BLOCK_SIZE="131072" TYPE="squashfs"  
/dev/loop11: BLOCK_SIZE="131072" TYPE="squashfs"  
/dev/loop2: BLOCK_SIZE="131072" TYPE="squashfs"  
/dev/fd0: SEC_TYPE="msdos" UUID="5C28-4B8D" BLOCK_SIZE="512" TYPE="vfat"  
/dev/loop0: BLOCK_SIZE="131072" TYPE="squashfs"  
/dev/loop9: BLOCK_SIZE="131072" TYPE="squashfs"  
/dev/loop16: BLOCK_SIZE="131072" TYPE="squashfs"  
/dev/loop7: BLOCK_SIZE="131072" TYPE="squashfs"  
/dev/sda2: UUID="cfed1901-89f6-42a9-a0f3-394a5e428f79" BLOCK_SIZE="4096" TYPE="ext4" PARTUUID="788387b0-dcdc-4fc2-b362-87f1399e207f"  
/dev/sda1: PARTUUID="a37361f1-ada6-41e8-8eba-a11d5b147259"  
/dev/loop14: BLOCK_SIZE="131072" TYPE="squashfs"  
/dev/loop5: BLOCK_SIZE="131072" TYPE="squashfs"  
/dev/loop12: BLOCK_SIZE="131072" TYPE="squashfs"  
/dev/loop3: BLOCK_SIZE="131072" TYPE="squashfs"  
/dev/loop10: BLOCK_SIZE="131072" TYPE="squashfs"  
jcsantos at JCSantos-Ubunut in ~  
└─○
```

14. Añadir una entrada de prueba en /etc/fstab (comentada)

Añade una línea:

```
# /dev/sdb1 /mnt/usbtest ext4 defaults 0 2
```

```
jcsantos@JCSantos-Ubunut:~  
jcsantos at JCSantos-Ubunut in ~  
└─○ sudo cat /etc/fstab  
# /etc/fstab: static file system information.  
#  
# Use 'blkid' to print the universally unique identifier for a  
# device; this may be used with UUID= as a more robust way to name devices  
# that works even if disks are added and removed. See fstab(5).  
#  
# <file system> <mount point> <type> <options> <dump> <pass>  
# / was on /dev/sda2 during curtin installation  
/dev/disk/by-uuid/cfed1901-89f6-42a9-a0f3-394a5e428f79 / ext4 defaults 0 1  
jcsantos at JCSantos-Ubunut in ~  
└─○ sudo nano /etc/fstab  
jcsantos at JCSantos-Ubunut in ~  
└─○ sudo cat /etc/fstab  
# /etc/fstab: static file system information.  
#  
# Use 'blkid' to print the universally unique identifier for a  
# device; this may be used with UUID= as a more robust way to name devices  
# that works even if disks are added and removed. See fstab(5).  
#  
# <file system> <mount point> <type> <options> <dump> <pass>  
# / was on /dev/sda2 during curtin installation  
/dev/disk/by-uuid/cfed1901-89f6-42a9-a0f3-394a5e428f79 / ext4 defaults 0 1  
# /dev/sdb1 /mnt/usbmount ext4 default 0 1  
jcsantos at JCSantos-Ubunut in ~  
└─○
```

15. Crear un archivo grande de forma artificial

`dd if=/dev/zero of=archivo_grande.img bs=1M count=100`

```
jcsantos@JCSantos-Ubunut:~  
jcsantos at JCSantos-Ubunut in ~  
└─○ dd if=/dev/zero of=ArchivoJC_Grande.img bs=1M count=100  
100+0 records in  
100+0 records out  
104857600 bytes (105 MB, 100 MiB) copied, 0.0431586 s, 2.4 GB/s  
jcsantos at JCSantos-Ubunut in ~  
└─○ ls -l  
total 102448  
-rw-rw-r-- 1 jcsantos jcsantos 104857600 Jan 19 13:51 ArchivoJC_Grande.img  
drwxr-xr-x 2 jcsantos jcsantos 4096 Jan 8 09:56 Desktop  
drwxr-xr-x 2 jcsantos jcsantos 4096 Jan 8 09:56 Documents  
drwxr-xr-x 2 jcsantos jcsantos 4096 Jan 8 09:56 Downloads  
drwxr-xr-x 2 jcsantos jcsantos 4096 Jan 8 09:56 Music  
drwxr-xr-x 2 jcsantos jcsantos 4096 Jan 8 09:56 Pictures  
drwxr-xr-x 2 jcsantos jcsantos 4096 Jan 8 09:56 Public  
drwxr-xr-x 2 jcsantos jcsantos 4096 Jan 8 09:56 Templates  
drwxr-xr-x 2 jcsantos jcsantos 4096 Jan 8 09:56 Videos  
-rwxrwxr-x 1 jcsantos jcsantos 1629 Jan 19 12:49 info.sh  
drwx----- 6 jcsantos jcsantos 4096 Jan 19 12:00 snap  
drwxrwxr-x 2 jcsantos jcsantos 4096 Jan 19 12:03 temp  
jcsantos at JCSantos-Ubunut in ~  
└─○
```

16. Ver cuánto espacio ocupa un directorio

`du -sh /var`

```
jcsantos@JCSantos-Ubunut:~  
jcsantos at JCSantos-Ubunut in ~  
└─○ du -sh /home/jcsantos  
212M /home/jcsantos  
jcsantos at JCSantos-Ubunut in ~  
└─○ sudo du -sh /var  
2.4G /var  
jcsantos at JCSantos-Ubunut in ~  
└─○
```

17. Ver espacio libre del disco

`df -h`

```
jcsantos@JCSantos-Ubunut:~  
jcsantos at JCSantos-Ubunut in ~  
└─○ df -Th  
Filesystem      Type      Size  Used Avail Use% Mounted on  
tmpfs           tmpfs     1.5G  2.1M  1.5G   1% /run  
/dev/sda2       ext4      20G   7.2G   12G  39% /  
tmpfs           tmpfs     3.7G   0  3.7G   0% /dev/shm  
tmpfs           tmpfs     5.0M   8.0K  5.0M   1% /run/lock  
tmpfs           tmpfs     3.7G   8.0K  3.7G   1% /tmp  
tmpfs           tmpfs     1.0M   0  1.0M   0% /run/credentials/systemd-journald.service  
tmpfs           tmpfs     1.0M   0  1.0M   0% /run/credentials/systemd-resolved.service  
tmpfs           tmpfs     739M  104K  739M   1% /run/user/1000  
jcsantos at JCSantos-Ubunut in ~  
└─○
```

18. Buscar archivos mayores de 100MB

```
find / -size +100M 2>/dev/null
```

```
jcsantos@JCSantos-Ubuntu:~  
jcsantos at JCSantos-Ubuntu in ~  
└─○ find / -size +100M 2>/dev/null  
/usr/lib/x86_64-linux-gnu/libLLVM.so.20.1  
/snap/gnome-42-2204/247/usr/lib/x86_64-linux-gnu/libLLVM.so.20.1  
/snap/gnome-42-2204/226/usr/lib/x86_64-linux-gnu/libLLVM-15.so.1  
/snap/firefox/7672/usr/lib/firefox/libxul.so  
/snap/firefox/6966/usr/lib/firefox/libxul.so  
/proc/kcore  
/sys/devices/pci0000:00/0000:00:0f.0/resource1  
/sys/devices/pci0000:00/0000:00:0f.0/resource1_wc  
/var/lib/snapd/seed/snaps/gnome-42-2204_226.snap  
/var/lib/snapd/seed/snaps/firefox_6966.snap  
/var/lib/snapd/snaps/gnome-42-2204_226.snap  
/var/lib/snapd/snaps/gnome-42-2204_247.snap  
/var/lib/snapd/snaps/firefox_7672.snap  
/var/lib/snapd/snaps/firefox_6966.snap  
jcsantos at JCSantos-Ubuntu in ~  
└─○
```

19. Mostrar solo directorios dentro de /home

```
ls -d /home/*/
```

```
jcsantos@JCSantos-Ubuntu:~  
jcsantos at JCSantos-Ubuntu in ~  
└─○ ls -d /home/*/  
/home/alumno/ /home/jcsantos/  
jcsantos at JCSantos-Ubuntu in ~  
└─○
```

20. Ver qué servicios escriben en /var/log

```
ls /var/log
```

```
jcsantos@JCSantos-Ubuntu:~  
jcsantos at JCSantos-Ubuntu in ~  
└─○ ls /var/log  
README  
alternatives.log cups fontconfig.log speech-dispatcher vmware-network.5.log  
appport.log cups-browsed gdm3 sssd vmware-network.log  
apt dist-upgrade gpu-manager.log syslog vmware-vmsvc-root.1.log  
auth.log dmesg hp sysstat vmware-vmsvc-root.2.log  
bootstrap.log dmesg.0 installer ubuntu-advantage-apt-hook.log vmware-vmsvc-root.3.log  
btmpt dmesg.1.gz journal unattended-upgrades vmware-vmsvc-root.log  
chrony dmesg.2.gz kern.log vmware-network.1.log vmware-vmtoolsd-jcsantos.log  
cloud-init-output.log dmesg.3.gz lastlog vmware-network.2.log vmware-vmtoolsd-root.log  
cloud-init.log dmesg.4.gz openvpn vmware-network.3.log vmware-vmusr-jcsantos.log  
dpkg.log private vmware-network.4.log wtmp
```